1 - Introduction to Python for AI

Artificial Intelligence Lab Manual - Clean Study Edition

Learning Objectives

- Use Python variables, collections, loops, functions, and NumPy arrays for small AI-oriented tasks.
- Load and inspect tabular data with pandas.
- Create simple plots that reveal patterns in numerical data.
- Write short, readable functions that can be reused in later AI labs.

Background

AI experiments usually combine data preparation, numerical computation, algorithmic logic, and visualization. This lab establishes the Python workflow used throughout the remaining manuals. Focus on understanding what each operation does rather than memorizing syntax.

Core Concepts

Concept	Meaning in this lab
Python collections	Lists, tuples, dictionaries, and sets store states, features, and labels.
NumPy	Vectorized numerical operations and array manipulation.
pandas	DataFrame-based loading, filtering, grouping, and summary statistics.
Matplotlib	Basic plots for inspecting distributions and relationships.
Functions	Reusable units that make later algorithm implementations easier to test.

Guided Procedure

1. Create a notebook and import numpy, pandas, and matplotlib.pyplot.

2. Generate a one-dimensional NumPy array of random integers and compute mean, median, minimum, and maximum.

3. Convert the array into a DataFrame with a descriptive column name.

4. Add a derived column, for example a normalized value or a category based on a threshold.

5. Plot a histogram and a line plot. Explain what each plot reveals.

6. Wrap one repeated transformation in a function and test it on two inputs.

Reference Implementation

Use the following as a starting point. Read and modify it rather than treating it as a final answer.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

rng = np.random.default_rng(7)
values = rng.integers(0, 100, size=30)

df = pd.DataFrame({"score": values})
df["above_mean"] = df["score"] > df["score"].mean()

print(df.describe())

plt.hist(df["score"], bins=8)
plt.xlabel("Score")
```

```
plt.ylabel("Frequency")
plt.show()
```

Lab Exercises

Task 1. Write a function that returns the largest and smallest values from a list without using max() or min().

Task 2. Create a DataFrame containing five synthetic students and three numeric features. Add a fourth feature derived from the first three.

Task 3. Generate 100 random values and compare their mean and median. Create a histogram and explain any difference.

Task 4. Filter a DataFrame using two conditions and report how many rows remain.

Task 5. Extension: implement z-score normalization from the formula rather than using a library function.

Suggested Deliverables

- Notebook or Python file containing the completed implementation.
- Short result table or output evidence for each required comparison.
- A concise interpretation of what changed and why; do not submit output without explanation.

Review Questions

1. Why are NumPy arrays generally preferred over Python lists for numerical work?

2. What is the difference between a DataFrame row filter and a column selection?

3. Why should data be visualized before training a model?

Completion check: You should be able to explain the algorithm or workflow in your own words, reproduce the main result, and identify at least one limitation.